

Skills Summary

Postulates, Theorems, Converses and Biconditionals

Use this reference while completing your worksheet or when studying.

The four labels. **Definition** — states what a term means; reversible by agreement.
Postulate — assumed without proof, so the system has a starting point. **Theorem** — proved from definitions, postulates and earlier theorems. **Biconditional** — “ p if and only if q ”, valid only when $p \rightarrow q$ and $q \rightarrow p$ are *both* true.
The converse of “if p , then q ” is “if q , then p ”. Swapping is free; *truth is not carried along*. One counterexample kills a converse.

1. Applying a postulate to compute

Two postulates do most of the algebra in this unit.
Segment Addition: if B is between A and C , then $AB + BC = AC$.
Angle Addition: if D is interior to $\angle ABC$, then $m\angle ABD + m\angle DBC = m\angle ABC$.
 Name the postulate first, then write the equation it licenses.

Example: B is between A and C , $AB = x + 4$, $BC = 2x$, $AC = 22$. Find x .
Step 1. Betweenness is what licenses adding the parts, so the Segment Addition Postulate supplies the equation before any algebra happens.
Step 2. $(x + 4) + 2x = 22$ gives $3x + 4 = 22$, so $3x = 18$.

$$x = 6 \quad (\text{check: } 10 + 12 = 22)$$

Try it: $\overrightarrow{BD}$ is interior to $\angle ABC$ with $m\angle ABD = 3x$, $m\angle DBC = 2x + 10$, $m\angle ABC = 85^\circ$. Find x .

check: $15 = x$

2. Postulate, theorem, or definition?

Ask two questions.
Could it be proved from something earlier? No $\rightarrow$ postulate. Yes $\rightarrow$ theorem.
Is it telling me what a word means, both ways? $\rightarrow$ definition.
 Obviousness decides nothing: “vertical angles are congruent” is obvious and is still a theorem, because it has a proof.

Example: Classify *through any three non-collinear points there is exactly one plane*.
Step 1. Start with the provability question, since that is the only thing that separates a postulate from a theorem.
Step 2. There is no earlier statement about planes to derive it from — it is one of the assumptions that gives the word *plane* its behaviour, and it is not unpacking a definition. So it is a **postulate**. (*Classification is read by a person, not computed — flagged for manual review.*)

Try it: Classify a *segment bisector* is a segment, ray or line that divides a segment into two congruent segments, and give the reason.

check: definition — it states what the term means, and it reverses

3. Testing the truth of a converse

To test “if q , then p ”, hunt for one case where q holds and p fails. With an equation, that means solving **completely**: if the solution set is bigger than the value named, the extra solution is the counterexample.

Example: Is the converse of “if $x = 7$, then $x^2 = 49$ ” true?

Step 1. The converse claims $x^2 = 49$ forces $x = 7$, so the test is whether 7 is the only number whose square is 49 — solve, do not substitute.

Step 2. $x^2 - 49 = 0$ factors as $(x - 7)(x + 7) = 0$, giving two solutions.

$$x = -7 \text{ or } x = 7$$

$x = -7$ satisfies the hypothesis and not the conclusion, so the converse is **false**.

Try it: Is the converse of “if $x = 5$, then $3x = 15$ ” true? Solve $3x = 15$ completely and decide.

check: $x = 5$ only, so the converse is true

4. When a biconditional is valid

“ p if and only if q ” is valid exactly when the two sides describe the **same set of cases**. Check the conditional and its converse separately; if either fails, the biconditional fails. Definitions always pass this test — that is what makes them definitions.

Example: Is “ $5x + 2 = 17$ if and only if $x = 3$ ” valid?

Step 1. Two sides, two directions — compare the solution sets rather than checking one substitution.

Step 2. $5x + 2 = 17$ gives $5x = 15$, whose solution set is a single value; and substituting that value returns 17, so the reverse direction holds too.

$$x = 3 \quad \text{— both directions hold, so the biconditional is valid.}$$

Try it: Is “ $x^2 - 2x - 8 = 0$ if and only if $x = 4$ ” valid? Solve completely first.

check: $x = 4$ or $x = -2$, so it is not valid